

Bank Customer Churn Analysis

Exploratory Data Analysis Report

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Dataset: Bank Customer Churn Dataset (Kaggle)

Total Records: 10,000 Customers

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1. Executive Summary

This report presents an exploratory data analysis of a bank's customer churn datasets containing 10,000 customers across three countries – France, Germany, and Spain.

The analysis reveals that approximately 20.4% of customers have churned. Key drivers of churn include age, gender, geography, and account activity. Customer aged 51-60, female customers, and those based in Germany show significantly higher churn rates. Inactive members are nearly twice as likely to churn compared to active members.

These findings suggest that targeted retention strategies focusing on older customers, female segments, and German market could significantly reduce churn.

2. Data Overview

2.1 Dataset Source

The dataset was obtained from Kaggle (Bank Customer Churn Dataset by Gaurav Topre). It contains 10,000 rows and 12 columns representing customers from France, Germany, and Spain.

2.2 Key Variable

- credit_score: Customer's credit score
- country: Customer's country of residence
- gender: customer's Gender
- age: Customer's age
- tenure: Number of years with the bank
- balance: Account balance
- num_products: Number of bank products used
- active_member: Whether customer is active
- salary: Estimated annual salary
- churn: Whether customer churned (1) or not (0)

2.3 Data Cleaning Steps

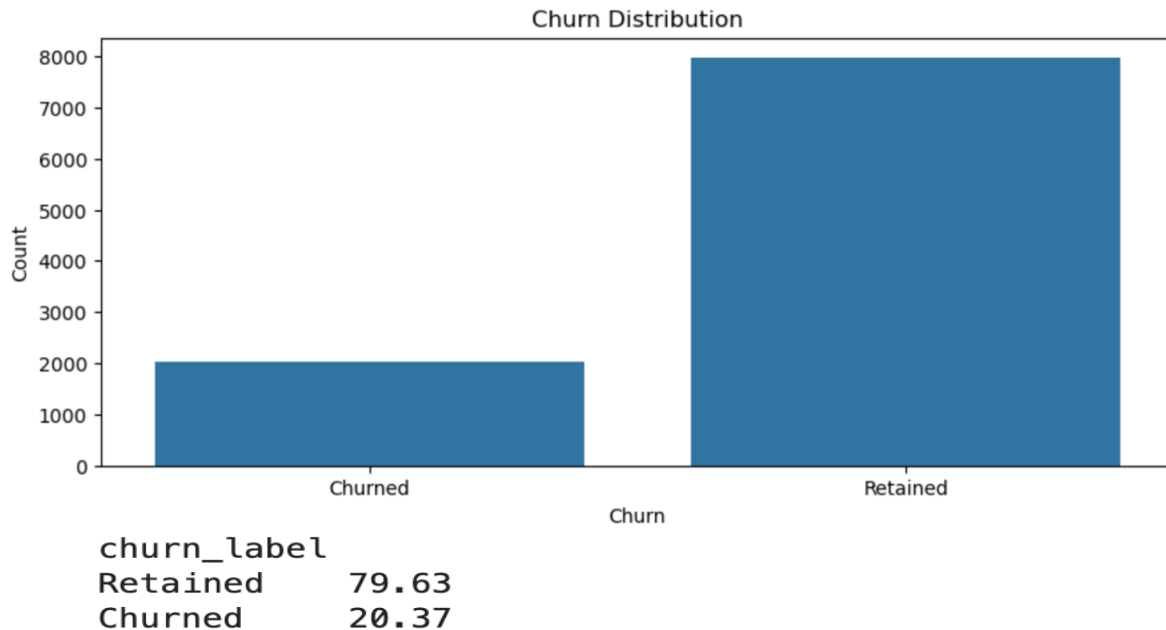
- No missing values found
- No duplicate records found
- Dropped customer_id column (not required for analysis)
- Created label columns for churn, credit_card, active_member
- Created age_group and balance_group bins for grouping
- Created credit_score_group bins for analysis

3. Exploratory Data Analysis

3.1 Univariate Analysis

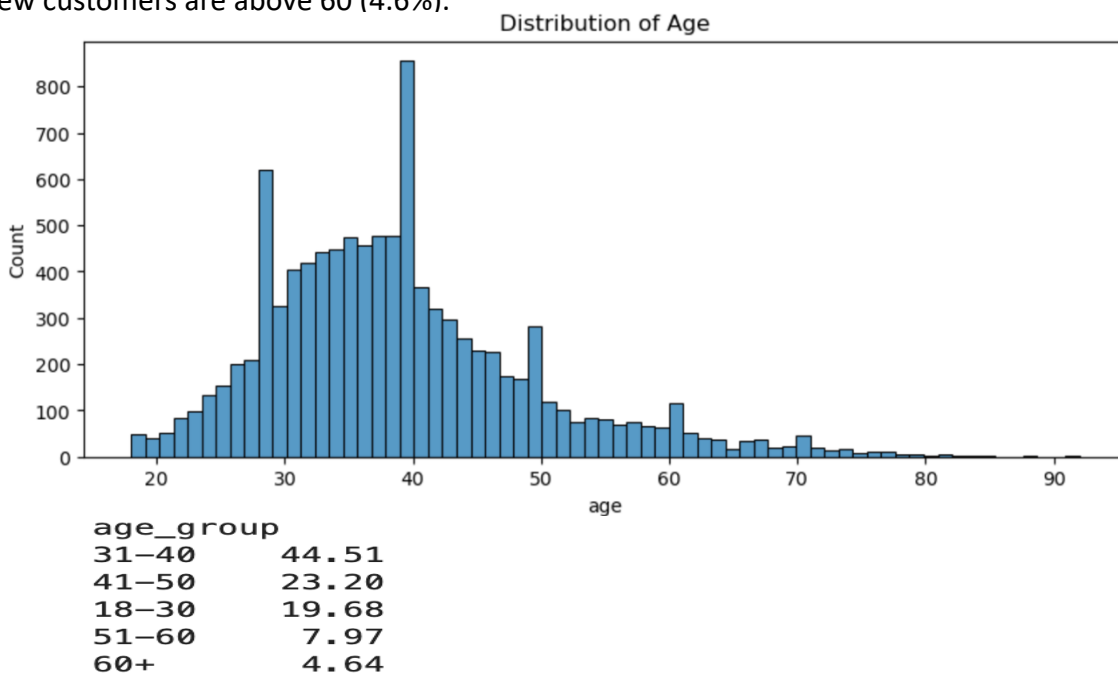
Churn Distribution

Approximately 20.4% of customer (2,037) have churned, while 79.6% (7,963) were retained. This indicates a class imbalance in the dataset.



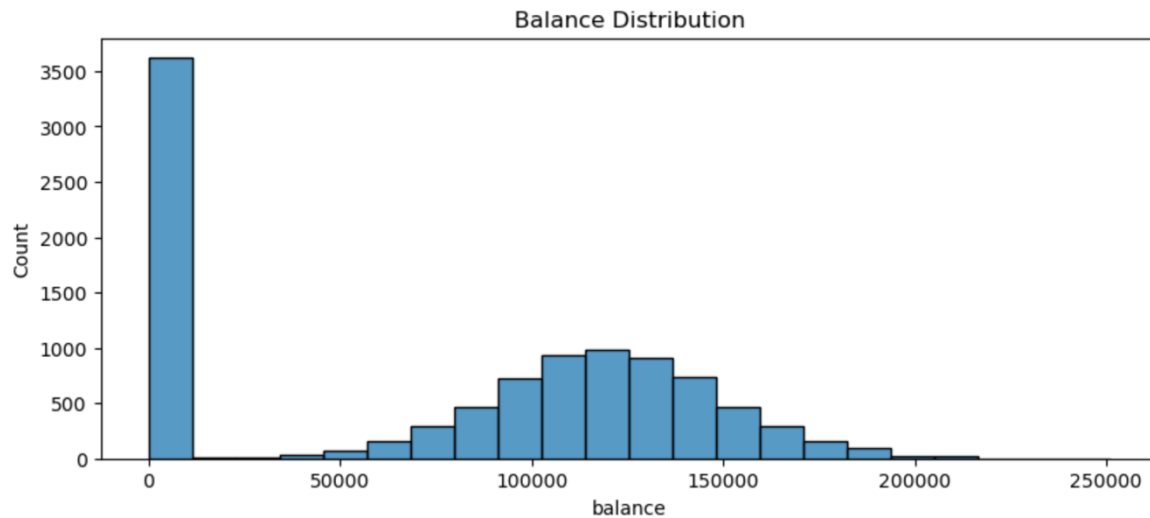
Age Distribution

The majority of customers fall in the 31-40 age group (44.5%), followed by 41-50 (23.2%). Very few customers are above 60 (4.6%).



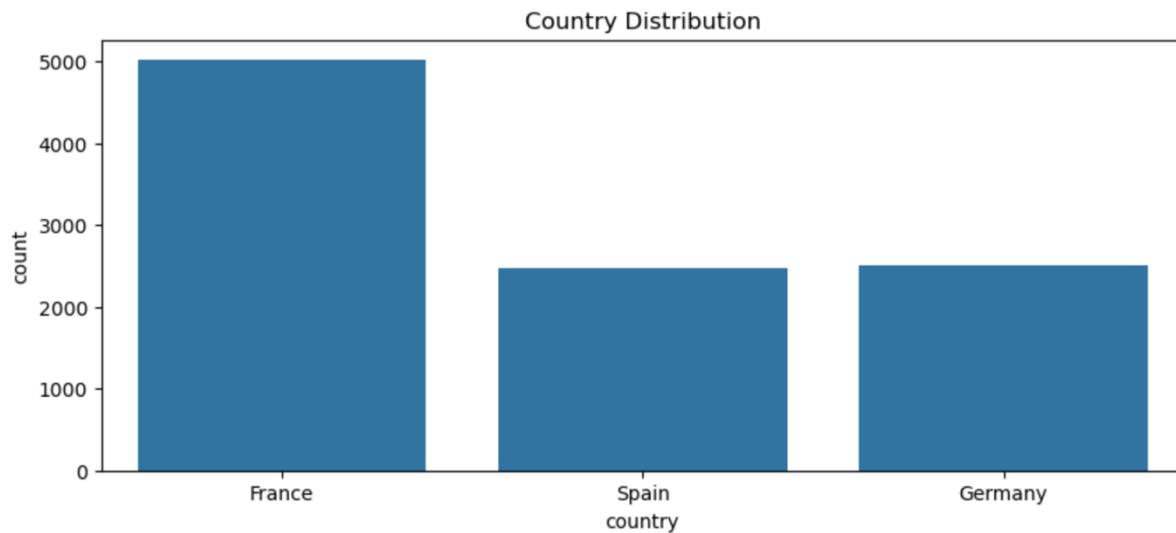
Balance Distribution

Balance distribution is bimodal - a large portion of customers have zero balance, while the remaining cluster around 100k - 150k. This suggests two distinct customer segments.



Country Distribution

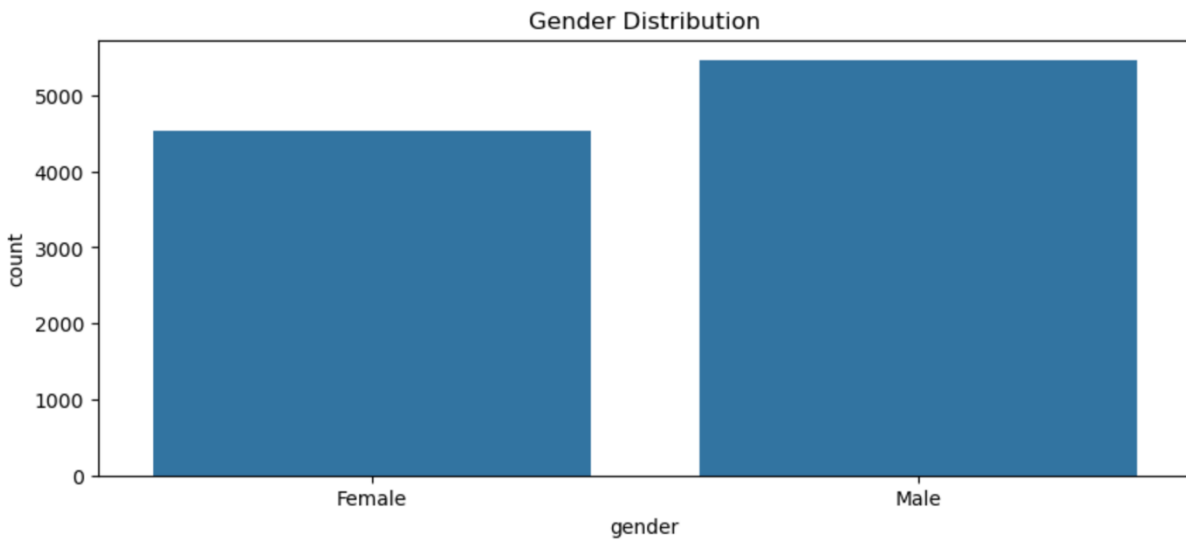
France has the highest customer base (50.1%), followed by Germany (25.1%) and Spain (24.8%).



country	
France	50.14
Germany	25.09
Spain	24.77

Gender Distribution

Male customers represent 54.6% while Female customers represent 45.4% of total customers.

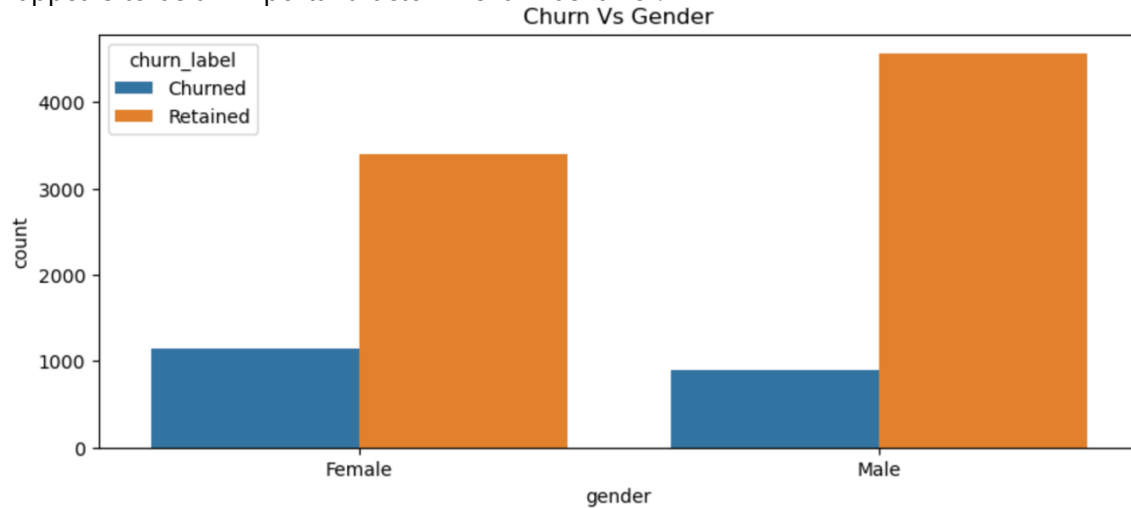


gender	
Male	54.57
Female	45.43

3.2 Bivariate Analysis

Churn by Gender

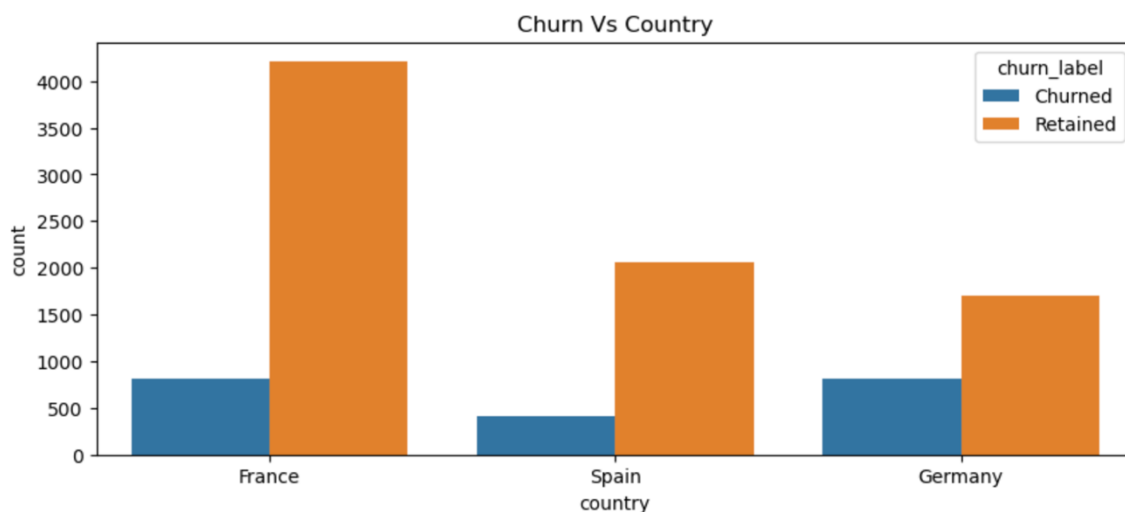
Female customers churn significantly more (25%) compared to male customer (16.5%). Gender appears to be an important factor in churn behavior.



gender	
Female	25.071539
Male	16.455928

Churn by Country

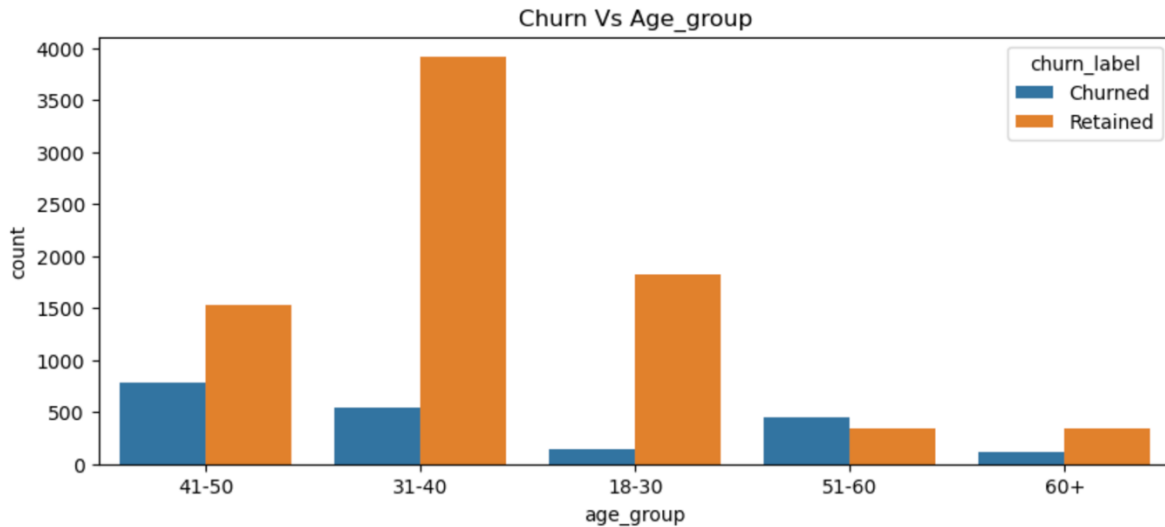
Germany has significantly high churn rate (32.5%) compared to Spain (16.7%) and France (16.2%). Germany customers are twice as likely to churn - needs business attention.



country	
France	16.154767
Germany	32.443204
Spain	16.673395

Churn by Age Group

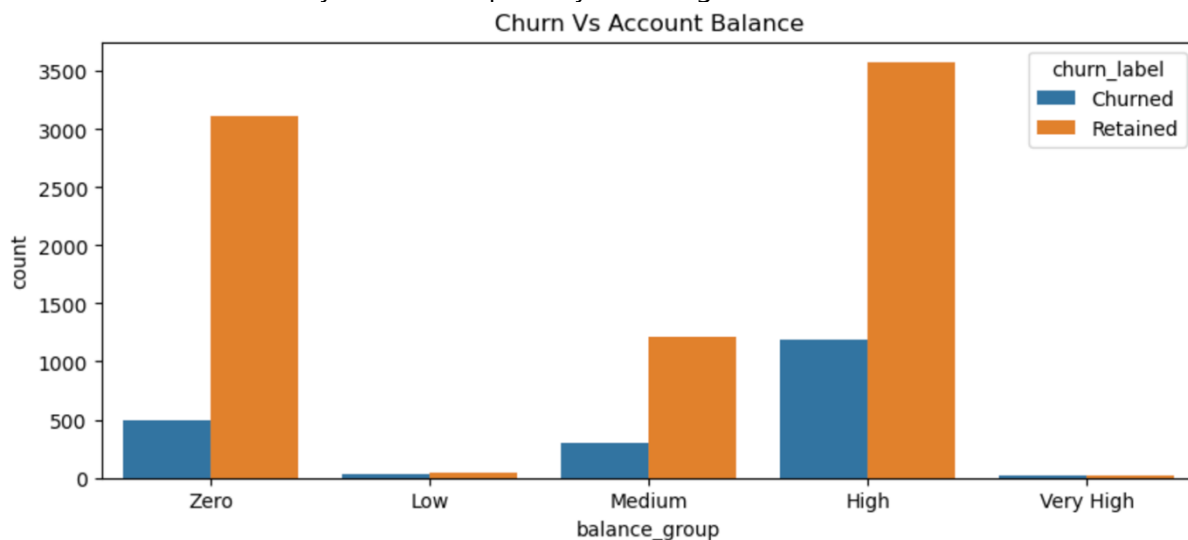
Churn rate increase significantly with age. 51-60 age group has the highest churn rate (56.2%), followed by 41-50 (34%). Younger customers (18-30) are least likely to churn (7.5%). Age is a strong predictor of churn behavior.



age_group	
18-30	7.520325
31-40	12.087171
41-50	33.965517
51-60	56.210790
60+	24.784483

Churn by Account Balance

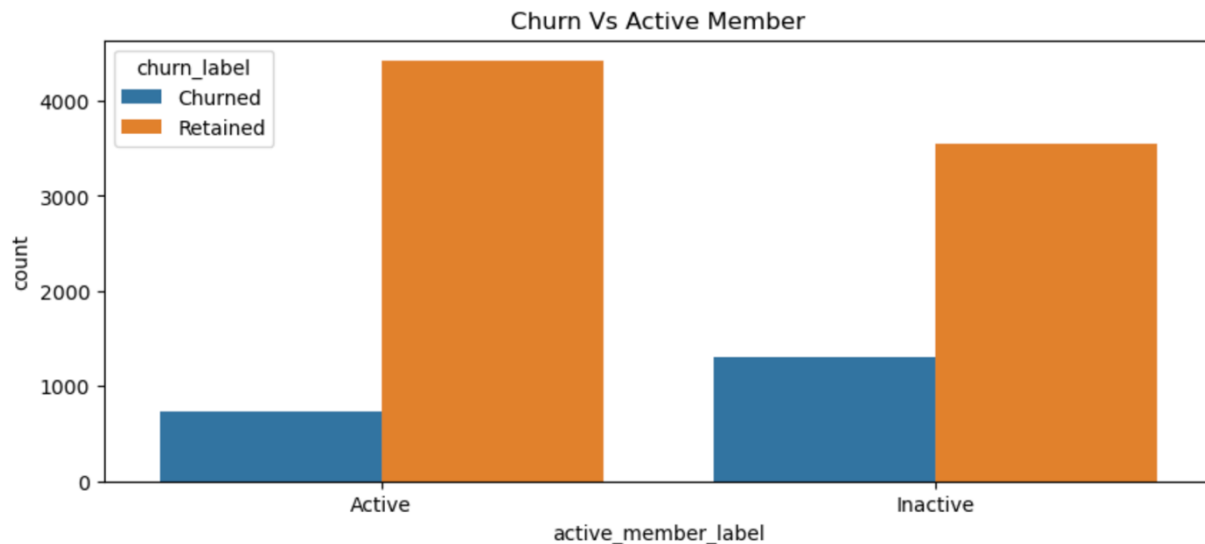
High balance customers churn significantly more (25%) compared to Zero balance (13.8%). Very High balance shows 55.9% churn but sample size is too small to draw conclusions. Customers with money in account are more likely to churn — possibly switching to better offers elsewhere.



balance_group	
High	25.015740
Low	34.666667
Medium	19.880716
Very High	55.882353
Zero	13.823611

Churn by Member Status

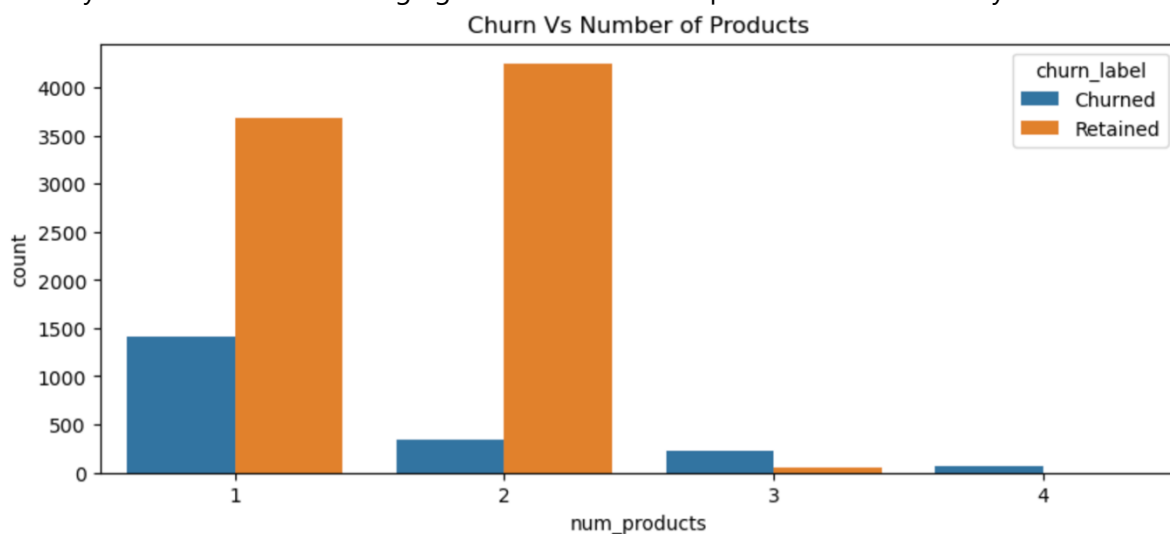
Inactive customers churn nearly twice as much (26.9%) compare to active customer (14.3%). Keeping customers engaged is critical to reducing churn.



```
active_member_label
Active      14.269074
Inactive    26.850897
```

Churn by Number of Products

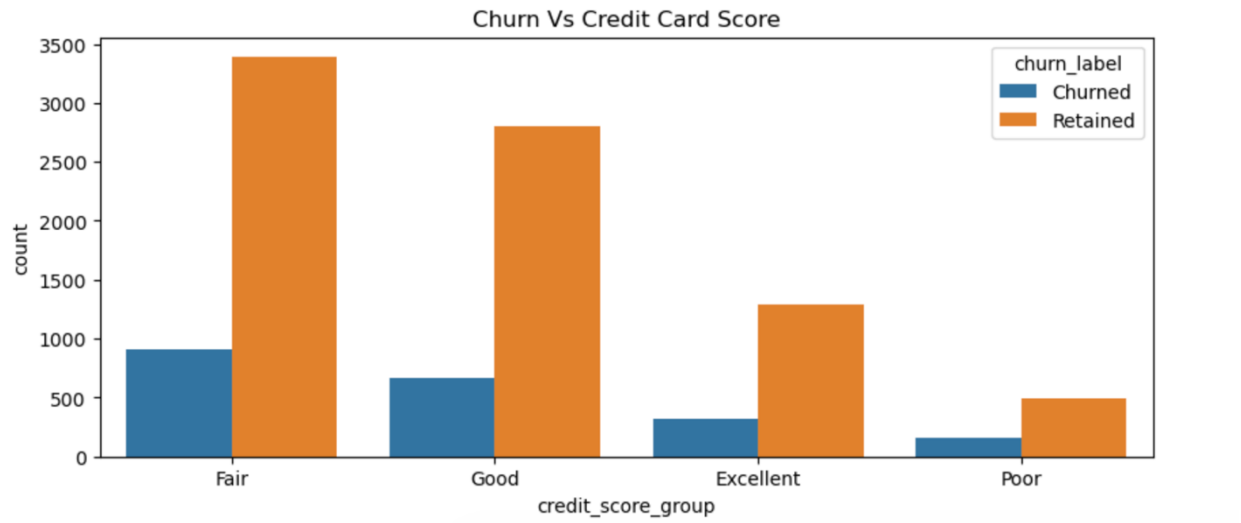
Customers with 1 product churn at 27.7%, while 2 product customers are most loyal (7.6%). Customers with 3-4 products churn extremely high (83-100%) but sample size is very small — results may not be reliable. Encouraging customers to take 2 products could be a key retention strategy.



```
num_products
1      27.714398
2       7.581699
3     82.706767
4    100.000000
```

Churn by Credit Score

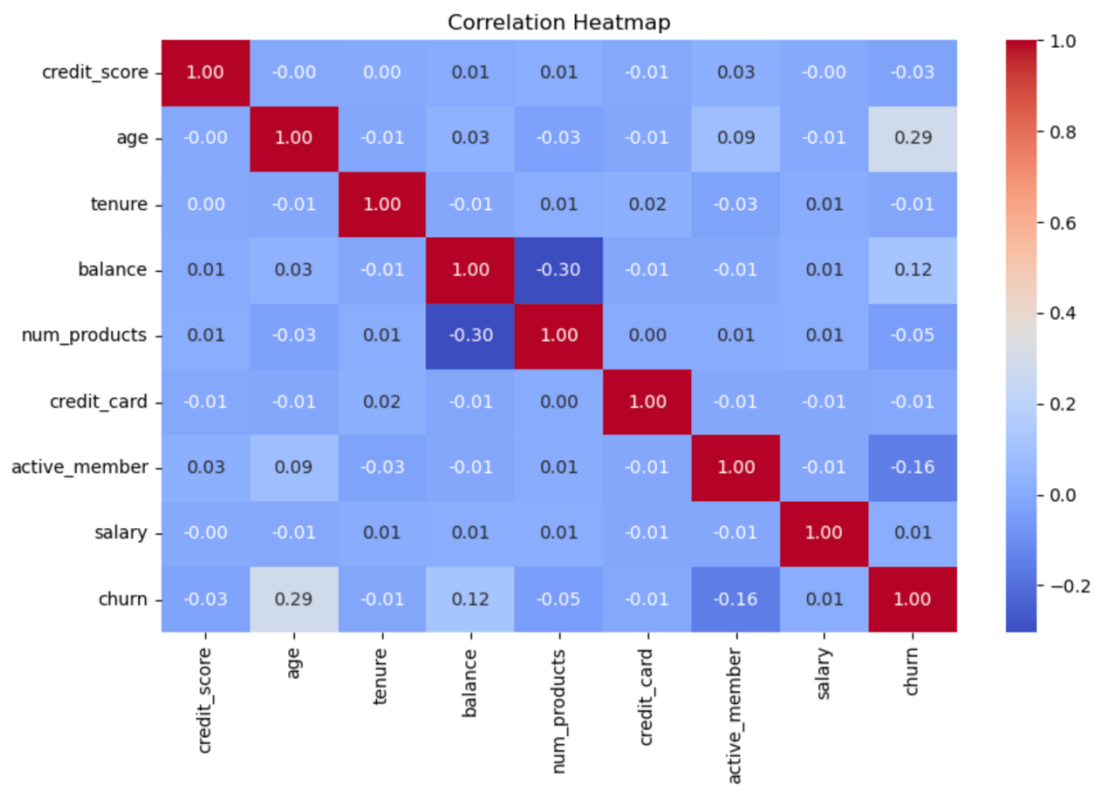
Poor credit score customers churn slightly more (23.6%) compared to Excellent (19.6%). However, the difference is small — credit score is not a strong predictor of churn.



credit_score_group	
Excellent	19.586984
Fair	21.075920
Good	19.249639
Poor	23.639191

3.3 Correlation Analysis

- Age has the strongest positive correlation with churn (0.29) — older customers churn more.
- Active member has negative correlation with churn (-0.16) — active members churn less.
- Balance & num_products are negatively correlated (-0.30) — more products mean lower balance.
- Salary and credit_score have almost no correlation with churn — confirming earlier findings.



4. Key Insights & Recommendations

4.1 Key Insights

1. Overall churn rate is 20.4% — 1 in 5 customers is leaving the bank.
2. Germany has double the churn rate compared to France and Spain (32.4% vs ~16%).
3. Female customers churn significantly more than male customers (25.1% vs 16.5%).
4. Older customers (51-60) are most at risk with a churn rate of 56.2%.
5. Inactive members are nearly twice as likely to churn (26.9% vs 14.3%).
6. Customers with 2 products are the most loyal (7.6% churn rate).
7. Salary and credit score have minimal impact on churn behavior.

4.2 Recommendations

1. Focus retention efforts on German market — investigate why churn is significantly higher and offer targeted incentives.
2. Design special retention programs for female customers — personalized offers, better customer service experience.
3. Engage older customers (41-60) with loyalty programs, premium services, or dedicated relationship managers.
4. Re-engage inactive members through email campaigns, app notifications, or exclusive offers before they churn.
5. Encourage customers to adopt a second product — customers with 2 products churn the least.

5. Conclusion

This exploratory data analysis of the Bank Customer Churn dataset reveals several critical factors driving customer attrition. With an overall churn rate of 20.4%, the bank faces a significant retention challenge.

The analysis identifies age, geography, gender, and member activity status as the strongest predictors of churn. Customers aged 51-60, female customers, German market customers, and inactive members represent the highest risk segments.

On the other hand, customers with 2 bank products show the highest loyalty, suggesting that cross-selling strategies could be an effective retention tool.

The insights derived from this analysis can guide the bank's marketing and customer success teams to design targeted, data-driven retention strategies — ultimately reducing churn and improving long-term customer lifetime value.

A Tableau dashboard has been developed alongside this report to enable interactive exploration of the data across different customer segments and geographies.